

验证报告

编号: NGBEC25003952302

日期: 2025 年 08 月 25 日

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客户名称: 浙江利尔达物联网技术有限公司

客户地址: 浙江省杭州市余杭区仓前街道文一西路 1326 号 1 号楼 1402 室

样品名称: NT26-FEU

测试基本型号: NT26FEUD60NNA

以上样品及信息由客户提供。

SGS 工作编号: NBP25-004645

样品接收时间: 2025 年 06 月 18 日

验证周期: 2025 年 06 月 18 日 ~ 2025 年 08 月 25 日

验证要求: 参考欧盟 RoHS 指令 2011/65/EU 的修正指令(EU) 2015/863。

验证方法: 见后续页。

验证程序和结果: 见后续页。

检测结果总结:

检测项目	结论
欧盟 RoHS 指令 2011/65/EU 附录 II 的修正指令(EU) 2015/863-铅、汞、镉、六价铬、多溴联苯 (PBB)、多溴二苯醚 (PBDE)、邻苯二甲酸二(2-乙基己基)酯 (DEHP)、邻苯二甲酸丁苄酯 (BBP)、邻苯二甲酸二丁酯 (DBP)和邻苯二甲酸二异丁酯 (DIBP)	符合

通标标准技术服务有限公司宁波分公司
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李纪成

Damon Li 李纪成
批准签署人

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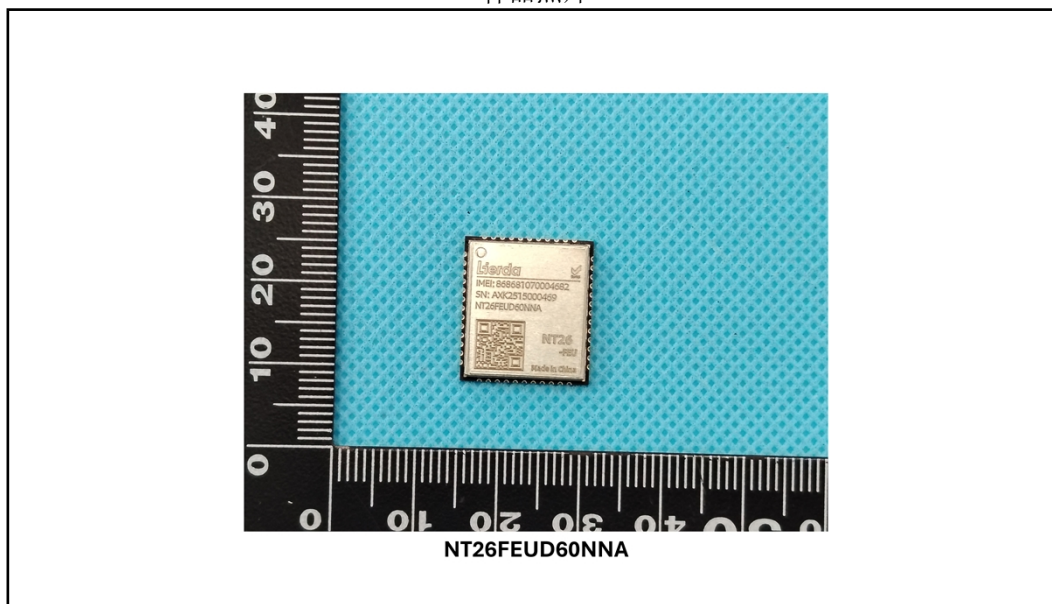
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样品照片



验证方法:

1. 参考 IEC 62321-2:2021, 拆分申请者提供的成品。
2. 参考 IEC 62321-1:2013, 仅对本报告照片中标识的样品进行了下列测试分析。
 - (1) 参考 IEC 62321-3-1: 2013, 用能量色散型 X 射线荧光分析仪器进行筛选。
 - (2) 湿法化学检测方法: 参考 IEC 62321-4:2013+A1:2017, IEC 62321-5:2013, IEC 62321-7-1:2015, IEC 62321-7-2:2017, IEC 62321-12:2023, IEC 62321-6:2015 和 IEC 62321-8:2017, 采用 ICP-OES, UV-Vis 和 GC- MS 进行分析。



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验证部件外观描述:

样品序号	样品编号	SGS 样品 ID	样品描述
SN1	A1	NGB25-0039523-0001.C001	银色金属片
SN2	A2	NGB25-0039523-0001.C002	黑色 PCB 本体
SN3	A3	NGB25-0039523-0001.C003	黑色 IC 本体
SN4	A4	NGB25-0039523-0001.C004	黑色 IC 本体
SN5	A5	NGB25-0039523-0001.C005	黑色电子元器件
SN6	A6	NGB25-0039523-0001.C006	银色金属焊锡
SN7	A7	NGB25-0039523-0001.C007	银色电子元器件
SN8	A8.1	NGB25-0039523-0001.C008.C001	黑色电子元器件 (A8 整改样提交于 2025-8-12)
SN9	A9	NGB25-0039523-0001.C009	黑色 IC 本体



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验证结果:

根据物料风险评估的结果, 验证送测样品中的以下材料。(除特别标注, 单位均为: mg/kg)

检测项目	A1	A2	A3	A4	A5	A6	A7	A8.1
Pb	BL	BL	BL	BL	BL	BL	BL	BL
Hg	BL	BL	BL	BL	BL	BL	BL	BL
Cd	BL	BL	BL	BL	BL	BL	BL	ND
Cr(VI)▼	BL	BL	BL	BL	ND	BL	ND	BL
PBB	---	ND	ND	ND	ND	---	ND	BL
PBDE	---	ND	ND	ND	ND	---	ND	BL
DEHP	---	ND	ND	ND	ND	---	ND	ND
BBP	---	ND	ND	ND	ND	---	ND	ND
DBP	---	ND	ND	ND	ND	---	ND	ND
DIBP	---	ND	ND	ND	ND	---	ND	ND
结论	符合	符合	符合	符合	符合	符合	符合	符合

检测项目	A9
Pb	BL
Hg	BL
Cd	BL
Cr(VI)▼	BL
PBB	BL
PBDE	BL
DEHP	ND
BBP	ND
DBP	ND
DIBP	ND
结论	符合

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备注:

(1) X 射线荧光光谱法(XRF)筛选结果说明:

(a)下表为 IEC62321-3-1: 2013 附录 A 不同基体材料中以 mg/kg 表示的筛选元素限值:

元素	聚合物	金属	复合材料
Cd	$BL \leq (70-3\sigma) < X < (130+3\sigma) \leq OL$	$BL \leq (70-3\sigma) < X < (130+3\sigma) \leq OL$	$LOD < X < (150+3\sigma) \leq OL$
Pb	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma) \leq OL$
Hg	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma) \leq OL$
Br	$BL \leq (300-3\sigma) < X$	不适用	$BL \leq (250-3\sigma) < X$
Cr	$BL \leq (700-3\sigma) < X$	$BL \leq (700-3\sigma) < X$	$BL \leq (500-3\sigma) < X$

(b) 现行法规限制的最大允许值是对 PBB/PBDE 和 Cr(VI) 而不是对 Br 和 Cr 的, 而 XRF 测定的是 Br 和 Cr。如果元素 Br 和或 Cr 的定量结果均高于限值(Br 计算基于在常见物质中 PBB/PBDE 的组成的理论计算), 则样品的筛选结果为“不确定”。

(c) EDXRF 的结果用于初步筛选, LOD=检出限, BL=低于限值, OL=超出限值, IN (符号“X”标记的区域)=不确定, 需要进一步研究, 我们建议客户用更精确的化学检测方法测试样品, 如用 ICP-OES 测定 Cd/Pb/Hg; 用 UV-Vis 测定 Cr(VI); 用 GC-MS 测定 PBB/PBDE。

(d) 对于复合材料, EDXRF 筛选测定元素所得数据可能与样品中实际含量不同。

(2) 邻苯二甲酸酯(PHTH)先进行筛分检测以确认样品是否符合限值要求, 符合则以“BL”表示, 若不符合则采用更精确的 GC-MS 化学检测方法测试样品。

检测项目	CAS No.
邻苯二甲酸二(2-乙基己基)酯 (DEHP)	117-81-7
邻苯二甲酸丁苄酯 (BBP)	85-68-7
邻苯二甲酸二丁酯 (DBP)	84-74-2
邻苯二甲酸二异丁酯 (DIBP)	84-69-5

(3) 化学检测方法结果说明:

(a) mg/kg = 0.0001%, MDL=方法检出限, ND =未检出(低于 MDL), --- =不适用。

(b) 化学检测方法的表达单位和方法检出限

检测项目	Pb	Hg	Cd	DEHP	BBP	DBP	DIBP
单位	mg/kg	mg/kg	mg/kg	mg/kg	mg/kg	mg/kg	mg/kg
方法检出限	10	10	10	100	100	100	100

任何一种 PBB 和 PBDE 的方法检出限是 100 mg/kg;

聚合物, 复合材料和皮革样品中 Cr(VI)的方法检出限是 10 mg/kg。

金属样品中 Cr(VI)的方法检出限是 0.10 $\mu\text{g}/\text{cm}^2$ 。

(c) ▼ = 金属样品中

a. 当六价格的浓度高于 0.13 $\mu\text{g}/\text{cm}^2$ 时, 样品为阳性, 即含有六价格;b. 当六价格的浓度为 ND(低于 0.10 $\mu\text{g}/\text{cm}^2$)时, 样品为阴性, 即未检测到六价格;c. 当六价格的浓度介于 0.10 $\mu\text{g}/\text{cm}^2$ 与 0.13 $\mu\text{g}/\text{cm}^2$ 之间时, 无法直接判定是否检测到六价格, 因不同个体的样品表面差异可能会影响测定结果;

由于未获知样品的存储条件和生产日期, 样品的六价格检测结果仅能代表检测时样品含六价格的状态。

(4) RoHS 指令中限制物质及均质材料中最大允许浓度值: Cd: 0.01%, Pb/Hg/Cr(VI)/PBB/PBDE/ DEHP /BBP/DBP/DIBP: 0.1%, 限值引用自 RoHS 指令(EU) 2015/863。



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(5) IEC 62321 系列等同于 EN 62321 系列。

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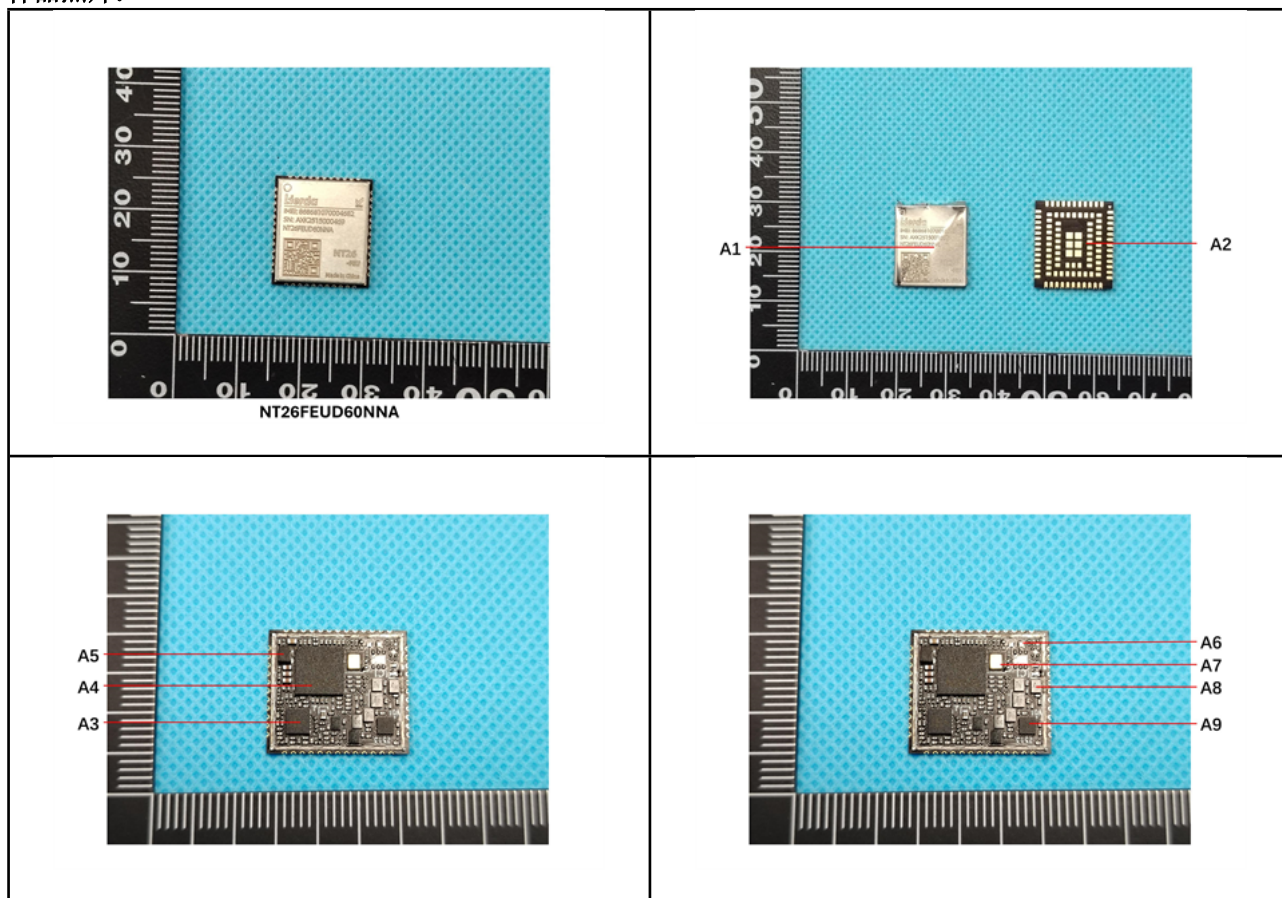
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